Arab Academy
for Science, Technology and Maritime Transport

Programming Fundamentals

WEEK #1

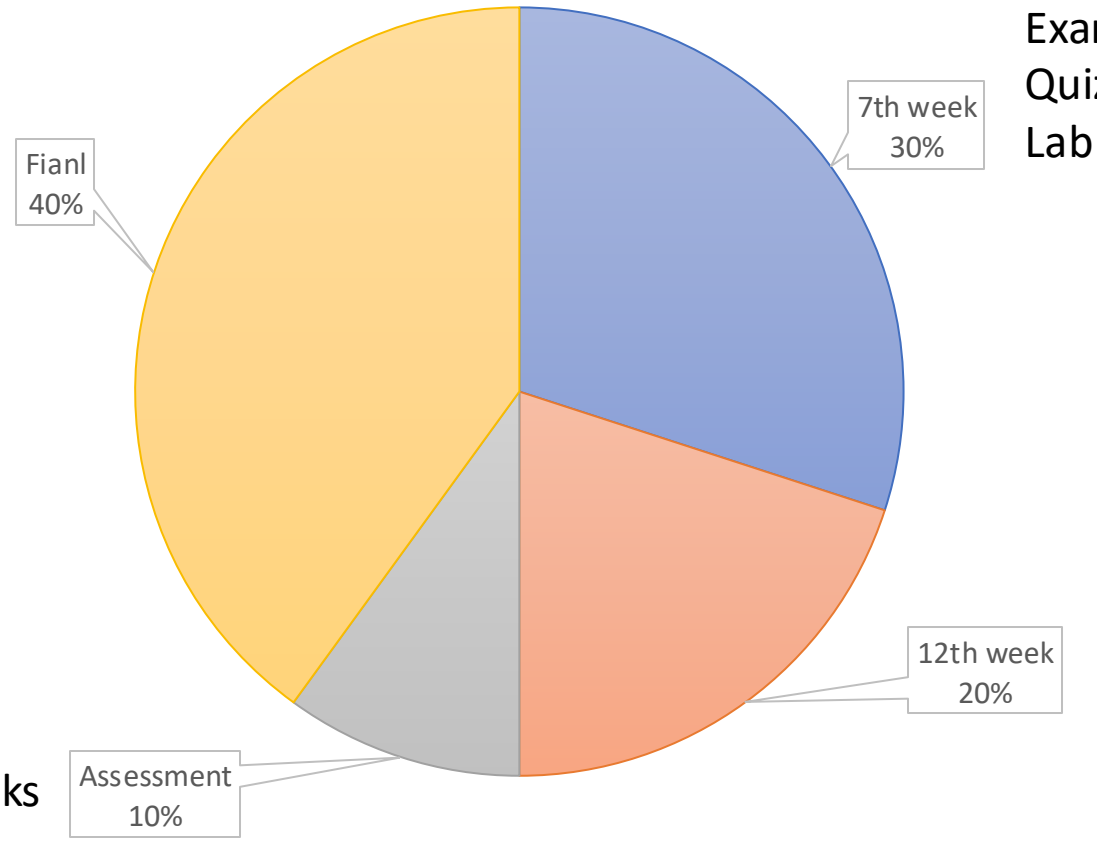
Lecture Agenda

- **Course Goals**
- **Course Grading System**
- **Course Timeline by weeks**
- **What is a Program?**
- **What is Programming?**
- **Programming Life-Cycle Phases**
- **Problem Solving**

Course Goals

- Introduce General concepts of Programming .
- Begin to Think Like a Programmer.
- Start Programming With C++ (a powerful, widely-used programming language).
- Become Junior C++ Programmer

Course Grading System

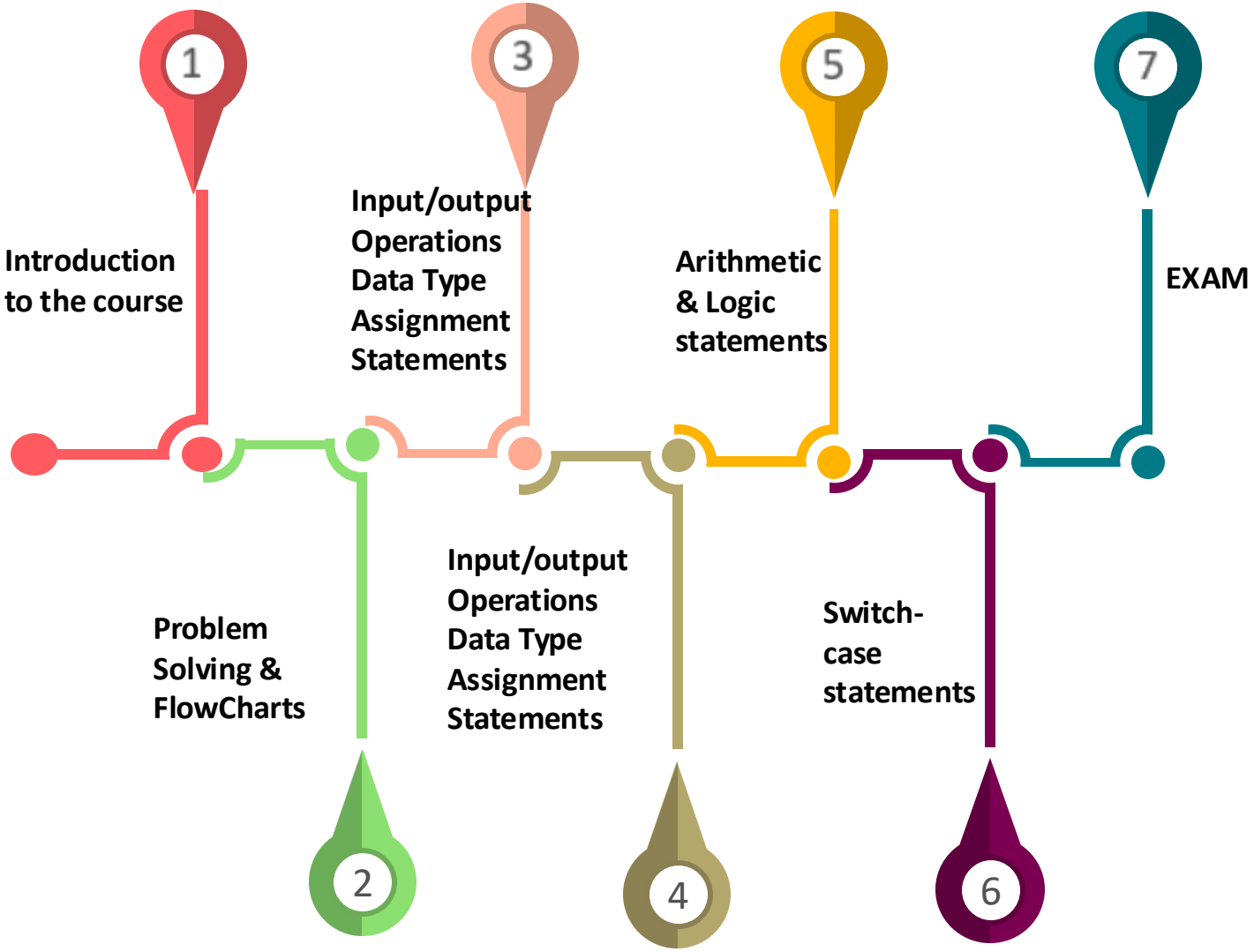


Exam→20 marks
Quiz-→5 marks
Lab work-→5 marks

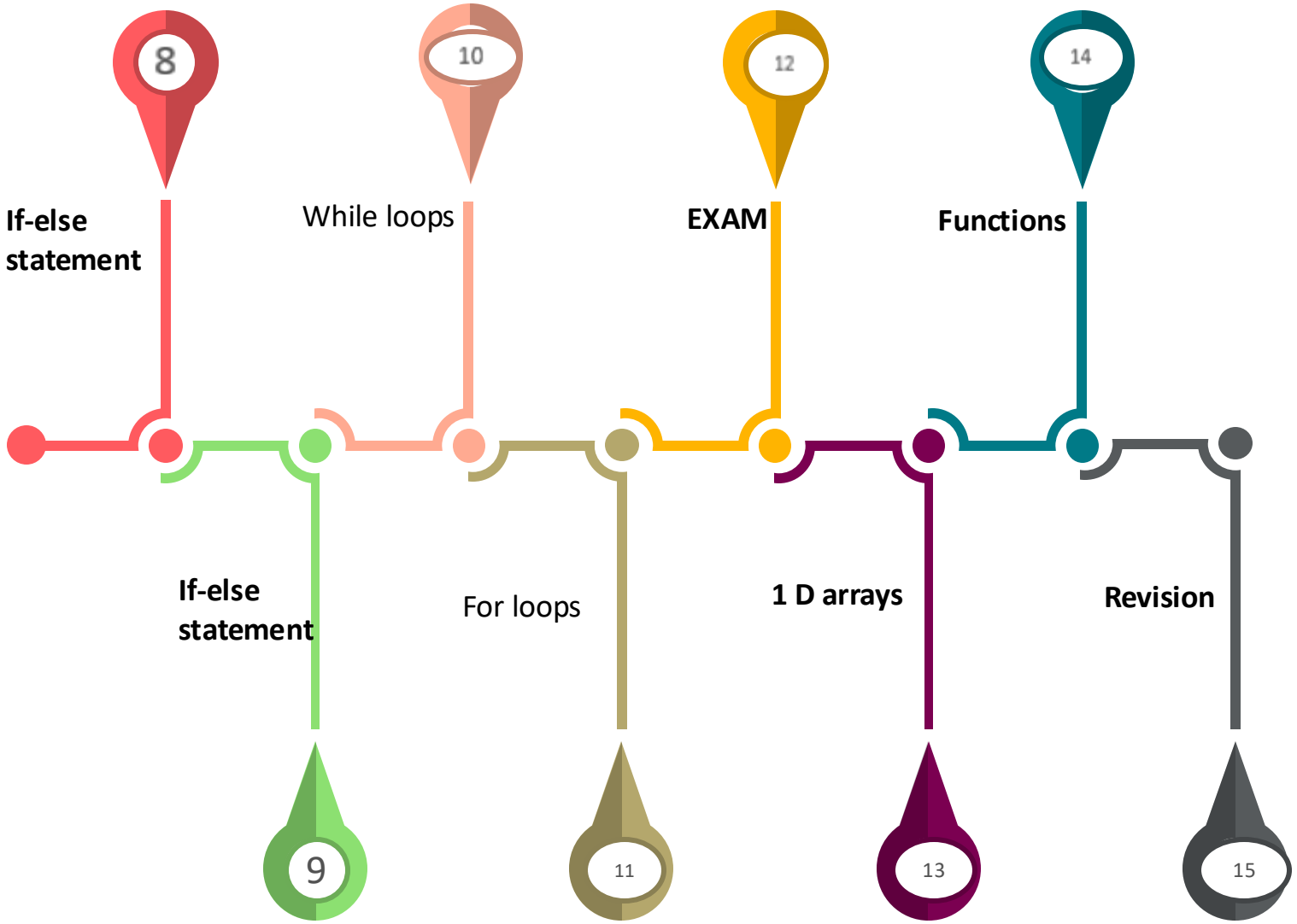
Exam→20 marks

Quiz-→5 marks
Lab work-→5 marks

Course Timeline by weeks



Course Timeline by weeks



What is a Program?

- Computers process data under the control of sets of instructions called computer programs.
- These programs guide the computer through ordered actions specified by people called computer programmers.
- The programs that run on a computer are referred to as software.

What is Programming?

- Find an algorithm to solve a problem.
- Programmers take this algorithm to write a program using any programming language.
- Can u give us some examples of programming languages??

Programming Life Cycle Phases

Programming Life Cycle Phases

<u>1- Problem Analysis :</u>

- During problem analysis, a systems analyst and programmer review specifications and talk with users to fully understand what the software should do.
- Documentation consists of:
  - Program specifications,
  - Timetable,
  - Which language will be used,
  - How the program will be tested,
  - What documentation is required.

Programming Life Cycle Phases

<u>2-Program Design :</u>

- This is the stage where program specifications are expanded into a complete design of the new program.
- The design approaches are either structure programming or object-oriented programming.
- The program design tools are the planning tools that includes:
  - Flow charts
  - Psuedocode

Programming Life Cycle Phases

<u>3-Program Coding :</u>

- Coding: actual process of creating the program in a programming language.
- Programming language must be chosen.
- Coding standards should be adhered to.
- Make use of reusable code and data dictionaries.
- Translate coded programs into executable code.

Programming Life Cycle Phases

<u>4-Program Debugging and Testing :</u>

- Debugging: process of making sure a program is free of errors or bugs. Preliminary bugging often finds syntax, logic or runtime errors.

- Testing: consists of alpha or beta testing.

Programming Life Cycle Phases

5-Program Maintenance:

- Program maintenance: process of updating software so that it continues to be useful.
- A costly process, but can be used to extend the life of a program

Problem Solving

- The single most important skill for a computer scientist is problem solving.
- Problem solving means the ability to formulate problems, think creatively about solutions, and express a solution clearly and accurately.

PROBLEM SOLVING

- Development of Computer Solution
    1. Identify or Define the problem
    2. Analyze the problem in terms of inputs, outputs, formulas, constants)
    3. Design the Solution
    4. Implement (program coding)
    5. Evaluate

PROBLEM SOLVING

<u>STEP #1: Identify or Define the problem</u>

It is required to calculate the sum of two numbers A and B and display result.

<u>STEP #2:Analyze the problem in terms of inputs, formulas, outputs and constants)</u>

Input(s): A,B

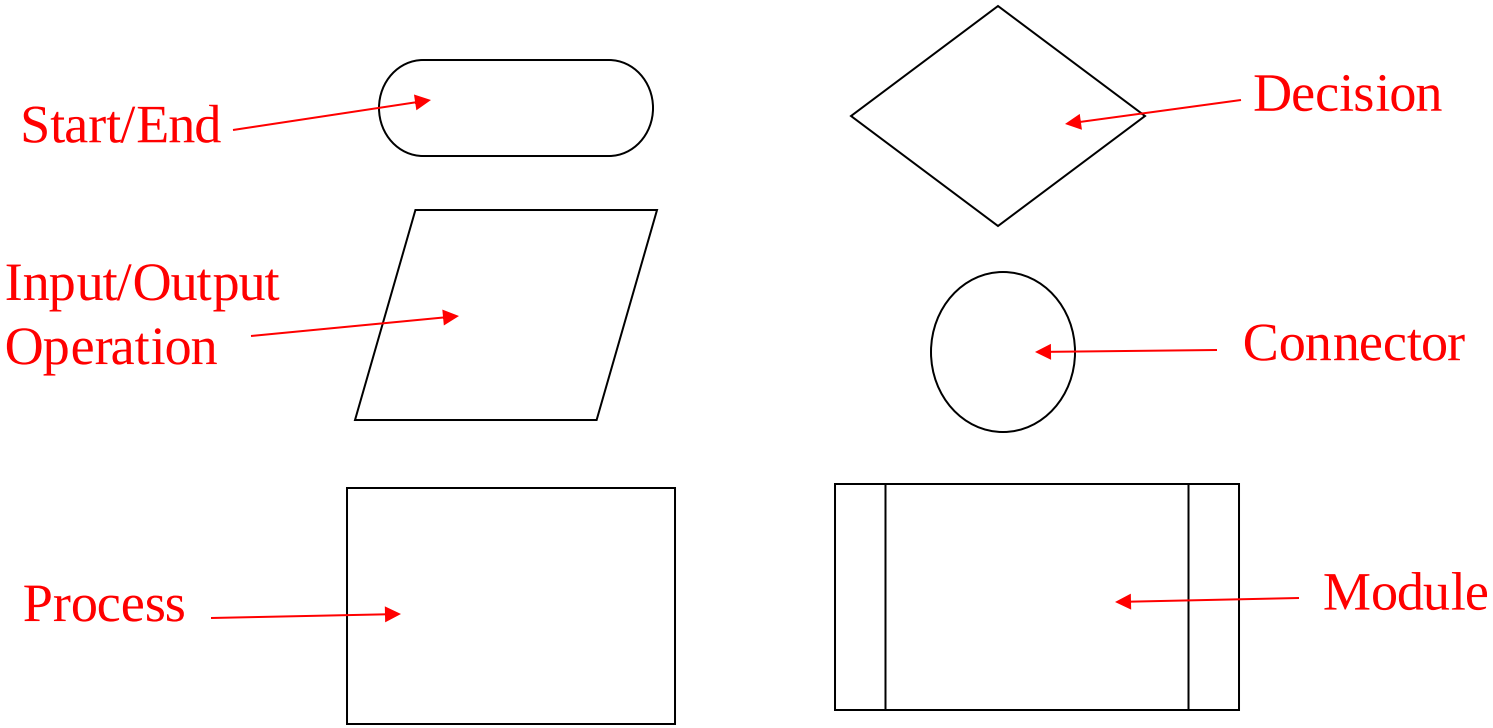
Formula(s):SUM=A+B

Output(s):SUM

PROBLEM SOLVING

STEP #3:Design the Solution

a) Flow Chart: is a diagram that depicts the “flow” of a program

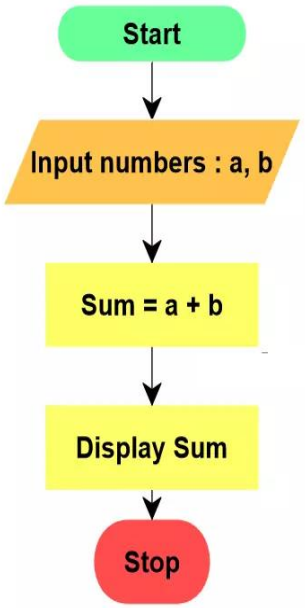


PROBLEM SOLVING

STEP #3:Design the Solution

b)Pseudocode: Uses English-like phrases terms to outline the program.

Flow chart



pseudo code

Step 1: Start.
Step 2 : Read A,B.
Step 3 : Sum = A + B.
Step 4 : Print Sum.
Step 5 : Stop.

PROBLEM SOLVING

STEP #4:Implement (program coding using C++)

```
#include <iostream>
using namespace std;
int main()
{  int a, b, sum;
   cout << "Enter two integers: ";
  cin >> a>> b;
  // sum of two numbers in stored in variable
  sum = a+ b;
  // prints sum
  cout << a<< " + " <<  b<< " = " << sum;
  return 0;
}
```

PROBLEM SOLVING

STEP #5:Evaluate

```
#include <iostream>
using namespace std;
int main()
{  int a, b, sum;
   cout << "Enter two integers: ";
  cin >> a>> b;
  // sum of two numbers in stored in variable
  sum = a+ b;
  // prints sum
  cout << a<< " + " <<  b<< " = " << sum;
  return 0;
}
```

RUN

```
Enter two integers: 4
5
4 + 5 = 9
```

Thank You

- Any Further Questions??